

PageRank: Random Surfer and the Google Matrix

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Motivation: Ranking nodes in a link graph

- We have a directed (or undirected) graph of pages/nodes and links/edges.
- A naive ranking (in-degree) ignores link quality.
- Idea: **a node is important if important nodes point to it.**
- This becomes a **Markov chain** over nodes (random walk on the graph).

Goal

Find a probability vector π (scores) such that π_i is the long-run fraction of time a random surfer spends at node i .

Random Surfer model (Markov chain)

Let A be the adjacency matrix and k_i the out-degree of node i . Define the random-walk transition matrix P (column-stochastic convention):

$$p_{ji} = \mathbb{P}(X_{t+1} = j \mid X_t = i) = \begin{cases} \frac{A_{ij}}{k_i}, & k_i > 0, \\ \frac{1}{N}, & k_i = 0 \text{ (dangling node)}. \end{cases}$$

- π is a stationary distribution if $\pi = P\pi$.
- Problem: in real graphs, P may be reducible/periodic $\Rightarrow$ no unique, stable ranking.

Teleportation and the Google matrix

To guarantee a unique ranking, introduce **teleportation**:

- with probability d follow a link (use P),
- with probability $1 - d$ jump to a random node (or personalized distribution).

Let v be a probability vector (uniform: $v = \frac{1}{N}\mathbf{1}$). Define the Google matrix:

$$G = dP + (1 - d) v\mathbf{1}^\top.$$

Then PageRank is the stationary distribution:

$$\pi = G\pi, \quad \sum_i \pi_i = 1, \quad \pi_i \geq 0$$

Why it works

For $0 < d < 1$, the chain is ergodic (unique stationary distribution; convergence from any start).

Computing PageRank (power iteration) + practical notes

Power iteration (dynamic method):

$$\pi^{(t+1)} = G \pi^{(t)}, \quad \pi^{(0)} = \frac{1}{N} \mathbf{1}.$$

Stop when $\|\pi^{(t+1)} - \pi^{(t)}\|_1 < \varepsilon$.

- Typical damping: $d = 0.85$.
- **Dangling nodes:** replace their column by v (uniform or personalized).
- **Personalized PageRank:** choose v non-uniform to bias toward a topic/user.
- Cost: each iteration is $O(|E|)$ if you use sparse matrices; usually converges fast.

Interpretation

π_i is the long-run probability that a random surfer is on node $i \Rightarrow$ a global importance score.

Why PageRank matters and how it is used

- **Importance from structure:** PageRank turns the link structure of a network into a robust global score.
- **Search and information retrieval:** helps rank web pages/documents beyond keyword matching by using connectivity.
- **Networks beyond the web:**
 - social networks (influence/authority ranking),
 - citation networks (paper/journal impact),
 - recommendation systems (item/user graphs),
 - biological networks (protein/gene interaction importance).
- **Personalization and control:** with a non-uniform teleportation vector v , we get *personalized PageRank* tailored to a user, topic, or trusted set of nodes.
- **Practical output:** sort nodes by π_i to obtain a ranking; use top- k nodes for retrieval, monitoring, or recommendations.